GUNJAN KUMAR MISHRA

AI ENGINEER | COMPUTER VISION & APPLIED AI

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience building and evaluating applied AI systems across computer vision, multimodal AI, model-serving APIs, and numerical forecasting. Develops Python/PyTorch, YOLO/DETR/SAM, vLLM, and FastAPI workflows on AWS, with strengths in benchmarking, error analysis, reproducibility, and sensor-data modelling. Currently pursuing an MSc in Financial Engineering.

TECHNICAL SKILLS

Machine Learning & Computer Vision: Python, PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, YOLO, DETR, RF-DETR, SAM, OpenCV, DepthAI, OCR, and time-series modelling

LLM/VLM, APIs & Cloud: OpenAI and Anthropic APIs, vLLM, structured outputs, tool/function calling, retrieval-augmented workflows, FastAPI, Uvicorn, Pydantic, REST APIs, AWS EC2/S3, and Streamlit

Engineering & Evaluation: SQL/PostgreSQL, Linux, Git, backend integration, experimental design, benchmarking, ablation studies, statistical evaluation, error and uncertainty analysis, and reproducibility

PROFESSIONAL EXPERIENCE

Research Engineer | Angelswing Inc. — Dec 2025 - Present

Remote | South Korea

- Built FastAPI/Uvicorn and DepthAI services for real-time OAK-D and Arducam video streams, exposing inference results, processed frames, depth, and spatial outputs to backend and interactive applications.
- Developed and benchmarked YOLO, DETR, RF-DETR, and SAM pipelines for drone and construction-site imagery across accuracy, latency, model size, robustness, and failure modes; integrated DETR with SAM to generate object-level masks for spatial analysis.
- Prototyped image-to-3D raycasting, geolocation, drone-coverage analysis, time-lapse stabilization, rangefinding, and photogrammetry-versus-LiDAR workflows.
- Deployed AI workloads and managed datasets and model artifacts on AWS EC2/S3; collaborated with backend teams on API integration, testing, refactoring, and technical documentation.

DSML Trainer (Part-time) | Logispark Technologies Pvt. Ltd. — Sep 2025 - Present

- Design and deliver project-based training in Python, data science, machine learning, model evaluation, and deployment.

Assistant Lecturer & Final-Year Project Reader | Herald College Kathmandu — Oct 2024 - Nov 2025

- Taught advanced Python, high-performance computing, and big-data analytics; evaluated final-year projects from literature review and methodology through implementation, documentation, and deployment.

Research Assistant | Pulchowk Campus, Tribhuvan University — Apr 2024 - Oct 2024

- Developed and compared Temporal Fusion Transformer, LSTM, Random Forest, and process-based forecasting models using 40 years of daily hydrometeorological data across eight stations; the strongest TFT achieved R2 0.88, RMSE 22.5 mm, and MAE 17.0 mm.
- Defined temporal splits, antecedent variables, evaluation metrics, and seasonal error analyses; documented underprediction of monsoon extremes and communicated assumptions, uncertainty, and limitations in an EarthArXiv preprint.

Earlier Experience: Data Engineer Associate Trainee, Fusemachines (Sep 2023 - Oct 2023) | Coding Instructor, Codingal & Mero Coding Classes (Mar 2023 - Apr 2024)

SELECTED AI SYSTEMS & PROJECTS

Language & Vision-Language Systems | *Applied AI engineering*

- Built multi-step text and image workflows using commercial LLM APIs and open-source models with schema-validated outputs, tool/function calls, retrieval, fallback handling, and FastAPI services.
- Served and evaluated open-source models with vLLM across output quality, latency, throughput, context behaviour, resource use, and deployment trade-offs.

3D Spatial Mapping & Image-to-World Geolocation | *Drone imagery, camera geometry, and LiDAR*

- Developed camera/LiDAR workflows that map image detections and selected pixels into 3D site models, returning spatial coordinates and geolocation attributes for operational analysis.
- Built drone-coverage heatmaps and traceable processing workflows to identify over-scanned and under-scanned regions from source frames and 3D outputs.

Distributed Acoustic Sensing Research | *Evidence-grounded sensor analytics*

- Developed an audit framework for chronology confounding, event-level evaluation, and claim boundaries across the Hollin Hill and Brienz slope-monitoring datasets.
- Designed a prespecified discovery/holdout analysis using 30 previously unused earthquakes for Cook Inlet submarine DAS records.

EDUCATION

Master of Science - Financial Engineering | WorldQuant University **Sep 2025 - Present**
Online

- Relevant study: Financial Data Science, Risk Management, Quantitative Methods, Machine Learning in Finance, Deep Learning in Finance, and Time-Series Forecasting.

B.E. - Electronics, Communication & Information Engineering | Tribhuvan University **2018 - 2024**
Pashchimanchal Campus | Pokhara, Nepal

- Awarded a full scholarship through entrance-examination merit. Final project: real-time ball detection and trajectory prediction using YOLOv9 and Kalman filtering.

SELECTED PUBLICATIONS

Attention-Based Deep Learning for Runoff Forecasting - *EarthArXiv preprint*

Typing with Emotions - *TechRxiv preprint*

GLoW YOLOv9 and DETR for Enhanced Billboard Detection - *manuscript*

From Sensor-Wise Accuracy to Episode-Wise Evidence - *DAS slope-monitoring manuscript*

CERTIFICATIONS & LANGUAGE

Certifications: Microsoft Certified: Azure AI Fundamentals | TensorFlow: Advanced Techniques | Machine Learning Specialization | AWS Fundamentals Specialization

Language: IELTS Academic - Overall Band 8.0